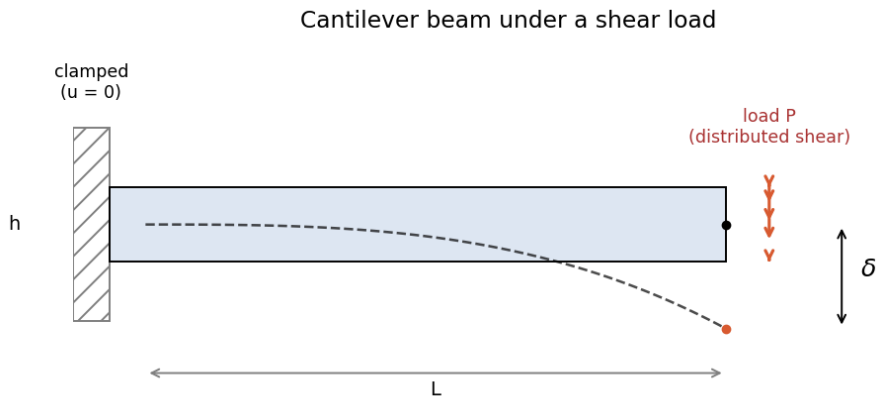


Assignment: The Cantilever Beam Problem
Linear Elasticity — Finite Element Method

1 Problem description

Consider a beam of length L and height h , clamped to a wall at one end and free at the other, like a diving board or a shelf bolted to a wall on one side only. A downward load is applied at the free end, and the beam bends. We want to compute how far the tip moves downward, and how the internal stress is distributed inside the beam.



The domain is the rectangle $\Omega = [0, L] \times [-h/2, h/2]$. The beam is **clamped** at $x = 0$: the material there cannot move at all, in any direction. The load is applied at $x = L$ as a **distributed shear force** over the height of the cross-section, not a point force: physically, a real force applied to a face of the beam acts over the whole face, and its exact distribution over that face is not uniform, as given in Section 4. The top and bottom edges, $y = \pm h/2$, are free: nothing is pushing on them.

Unlike a scalar problem such as the Poisson equation, where the unknown is a single number at each point (e.g. a temperature), here the unknown is a **vector**: at every point of the beam, the material is displaced by some amount in both the x and y directions. This displacement field is what determines both the shape the beam takes under load and the internal stress.

2 Governing equations

Strong form. Find the displacement field $\mathbf{u} = (u_x, u_y) : \Omega \rightarrow \mathbb{R}^2$ such that

$$\begin{cases} -\operatorname{div} \boldsymbol{\sigma}(\mathbf{u}) = \mathbf{0} & \text{in } \Omega \\ \boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\varepsilon}(\mathbf{u}), \quad \boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) & \text{in } \Omega \\ \mathbf{u} = \mathbf{0} & \text{on } \Gamma_D \ (x = 0) \\ \boldsymbol{\sigma} \mathbf{n} = (0, t_y(y)) & \text{on } \Gamma_N \ (x = L) \\ \boldsymbol{\sigma} \mathbf{n} = \mathbf{0} & \text{on } \Gamma_{\text{free}} \ (y = \pm h/2) \end{cases}$$

Constitutive law (plane strain, isotropic linear elasticity). In Voigt notation — $\boldsymbol{\varepsilon}, \boldsymbol{\sigma}$ packed as 3-vectors $(\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy})$, $(\sigma_{xx}, \sigma_{yy}, \sigma_{xy})$ — $\mathbf{C}$ is the 3×3 matrix:

$$\boldsymbol{\sigma} = \mathbf{C} \boldsymbol{\varepsilon}, \quad \mathbf{C} = \frac{E}{(1+\nu)(1-2\nu)} \begin{pmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{pmatrix}$$

Weak form. Find $\mathbf{u} \in V = \{\mathbf{v} \in H^1(\Omega)^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_D\}$ such that

$$\int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{u}) : \mathbf{C} : \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega = \int_{\Gamma_N} t_y v_y \, ds \quad \forall \mathbf{v} \in V$$

3 Task

1. Discretize the weak form with P1 (linear) triangular finite elements and assemble the linear system $Ku = F$.
2. Solve for u_h .
3. Compute the tip deflection $\delta_h = u_{h,y}(L, 0)$ and compare it to the Timoshenko reference solution δ (Section 4) at several mesh resolutions. Report the observed order of convergence.
4. Plot the deformed configuration and the von Mises stress field.

4 Data

Symbol	Value	Meaning
L	8	beam length
h	1	beam height
E	1	Young's modulus
ν	0.3	Poisson's ratio
P	1	load at the free end

$$I = \frac{h^3}{12}, \quad G = \frac{E}{2(1+\nu)}, \quad A_s = \frac{5}{6}h$$

$$t_y(y) = \frac{P}{2I} \left(\frac{h^2}{4} - y^2 \right), \quad y \in [-h/2, h/2]$$

$$\delta = \frac{PL^3}{3EI} + \frac{PL}{GA_s}$$